**Question #1: First Question (15 marks): Fill-in the table with the correct answer:**

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 | Q9 | Q10 | Q11 | Q12 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| C | B | B | B | B | D | A | D | C | A | A | B |

1. If the size of memory is 512 bytes, what will be the starting and ending offset addresses of it?

A. 00H to FFH  B. 000H to 1FFH  [✓ C. 0000H to FFFFH]  D. 000H to FFFH

2. What is the type of addressing mode used in instruction MOV AL, [BP+6]

A. Indirect  [✓ B. Based relative]  C. Direct Indexed  D. Direct

3. the word length of Intel 8086 microprocessor is:

A. 24  [✓ B. 16]  C. 20  10  D. 32

4. 5. If the offset address of X in data segment is equal to 118H, and DS=2000H, then the physical address of X =?

20000+118
20118

A. 21118 H  [✓ B. 20118H]  C. 21117 H  D. 31117H

5. Which is part of execution unit in Intel 8086 microprocessor?

A. Segments registers  [✓ B. ALU]  C. physical address generating unit  D. IP register

6. Show the binary value in register AL after execution the following instruction sequence,

```
MOV AL,11D → Ah = 11     0B
MOV BL,15D  Bh = 15 → 0F
SUB AL,BL  → Ah=0B − 0F →
```

A. AL=11111111  B. AL=11111011  C. AL=11111110  [✓ D. AL=11111100]

7. If the SP register contains the hexadecimal value F11FH, what will be its value after executing the following program segment?

```
PUSH AX
PUSH BX
ADD AX,BX
POP AX
POP AX
PUSH CX
```

[✓ A.F11FH]  B. F11AH  C. F11CH  D. F11DH

8. Find the value of AX after execution the following assembler codes

```
MOV CX,5    → CX=005
MOV AX,0      AX=0000
L: ADD AX,CX → ADD= 00+05 =0005    AX=0005 + 0004 = 0009 , AX=0002
DEC CX      → CX=0005-1 → 0004  , CX= 0004 - 0001 = 0003  CX= 0002
CMP CX,2    →   000" > 2                                    stop
JA L
HLT
```

A. 9  B. 15  C.8  [✓ D. 12]

9. Which is not part of execution unit in Intel 8086 microprocessor?

A. Segments registers  B. ALU  [✓ C. physical address generating unit]  D. IP register

10. The content of register BL after execution the instruction MOV BL,-11

[✓ A. BL=F6H]  B. BL=F7H  C. BL=0BHH  D. BL=06H

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11. Find the content of AL and zero flag (ZF) and carry flag (SF) after execution of the following code:

```
MOV AL, 0FDH
ADD AL, 3        → Ah= 00   CF=1
ADC AL, 0FEH     → Ah = 00+1 → 0   ZF=0   → SF=1
```

[✓ A. AL=FFH,ZF=0;SF=1] ~~B. AL=00H,ZF=0;SF=1~~ C. AL=FFH,ZF=0;SF=0 C. AL=02H,ZF=0;SF=1

12. The offset of a particular segment varies from ----------to----------

A.000h to FFFh [✓ B. 0000h to FFFFh] C. 00h to FFh D. 00000h to FFFFFh

33

## Question #2 (8 points)

Trace out the following program segments and find registers contents:

```
        X      X  X+1  X+2 X+3 X+4 X+5 X+6
ARR  DW  3333H, 4444H, 1111H, 2222H, 0FFFH

MOV BX,OFFSET ARR     Bx=03
MOV SI, 4  →  SI=4
MOV  DI,2      DI=2
MOV DL, [BX ][SI]   Dh= X+4   Dh=11h
MOV AL, [BX]  →  Ah= X  →  Ah=33
XCHG AL, DL   →

AL=  11h  ;  DL=  33h
```

```
MOV AL, 22H          AL=22h
SUB  AL,22           Ah= 22-22 = 0   ZF= zero
JZ YES  → ZF=1
SUB AL,5
JMP EE
YES: ADD AL,5    Ah= 0+5 = 05
EE:HLT

AL=  05h
```

```
ARRAY    DB   8, 3, 2, 4,6,8,5,7,0
MOV AL,0          Ah=0                 0A+06
MOV CX,4          CX=4                 0010
MOV BX, OFFSET ARRAY    Bx=C3          EA
L:                            08+02 = 
ADD  AL, [BX]   → Ah =  0+08 = 08 , 08+
ADD BX,2        →  Bx =  X+2  =  X+4, X+6
LOOP L   CX=3 , CX=2 , CX=1 , CX=0    0015
  MOV    AH,[BX]    →

AL= 15 ~~11~~  AH= 00h
```

08, 03, 02, 04, 06, 08, 05, 07, 00

```
X DW 104FH, 201AH, C030H
MOV BX, OFFSET  X      Bx= [X]
MOV AX, [BX]   → AX= 104F
INC  BX        → X+1
MOV   CX,[BX]  →  CX = X+1 = 1A10
PUSH AX    →  201A
PUSH CX    →  C030
POP AX     →  ~~201A~~ C030
    POP CX →  ~~104F~~ 201A

AX= ~~104F~~   CX= ~~1A10~~
AX= C030h   201Ah
```

4F, 10, 1A, 20, 30, C0

10, 4F, 20, 1A, CC, 30

## Question #3(10 points)

a) Suppose we have a microprocessor with 32bits address bus; what is the maximum memory capacity will be?

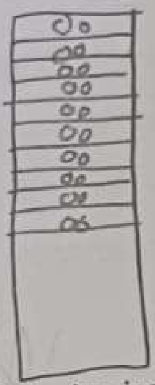
$$= 2^{32} = 2^{30} \cdot 2^{2} = 4GB$$

b) For the following instruction MOV [BX], 50H; what is the addressing mode used?

INdirect addresing

c) Write an assembler codes to store zeroes into X memory locations (3 points)

X DB 10 DUP (?)



d) Write the equivalent code of the following instructions (2 points)

LOOP L

= ADD CX

[illegible]

Jfz = 1

e) Write an assembler code to implement the following if statement: (3 points)

If X< 8 s=s+15 else s=s+7; assuming x and y are signed bytes.

RoG 100

HLT

Spark